KING KHALID UNIVERSITY ■ College of Computer Science

Genetic Mutation Detection Using Machine Learning

Presented by

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Problem statement: missense variants at clinical scale

Missense variants at clinical scale

- A point mutation that swaps one amino acid in a protein.
- Most common coding variant in the human genome.
- Most are benign; a clinically-important minority cause disease.
- Distinguishing the two is the central problem of clinical genetics.

Why prior work overstates performance

1. **Definitional circularity** — features derived from the label (e.g. `is_common`) inflate metrics.
2. **Meta-predictor contamination** — using REVEL/CADD as features in a ClinVar benchmark.
3. **Paralog leakage** — KRT1 in train + KRT14 in test $\Rightarrow$ 80% sequence identity $\Rightarrow$ model has seen most of the test.

Exome: ~20,000 variants/patient; human curation ~a few hundred/day. *Scale demands automation; automation demands honest evaluation.*

Research objectives + scope

Data + methodology

1. Assemble a paralog-disjoint corpus of 195k missense variants.
2. Eliminate three specific leakage sources.
3. Train and calibrate a defensible baseline.
4. Re-score published tools on our own test split.

Validation + delivery

5. Evaluate externally on DENOVO-DB.
6. Prove orthogonality of zero-shot ESM-2 signal.
7. Publish everything: code, tests, Docker, 10 narrative notebooks, Streamlit demo, academic report.

Every objective maps to at least one CI-enforced check.

Classical tools (2003–2010): conservation-based predictors

Tool (year)	Algorithm	Pros	Cons
SIFT (2003)	Conservation only — PSSM from PSI-BLAST	Instant via VEP REST; baseline of >95% clinical tools	No AA chemistry; arbitrary 0.05 threshold; weak on slow-evolving sites
PolyPhen-2 (2010)	Naïve Bayes on conservation + chemistry + 3D struct. (HumDiv)	Biochemical context (Grantham, BLOSUM); standard in clinical reports	HumDiv overlaps ClinVar $\Rightarrow$ contamination on ClinVar benchmarks

Where we re-score them

On our paralog-disjoint test (own-coverage): SIFT = 0.620 PR-AUC (96 % cov.), PolyPhen-2 = 0.728 (93 % cov.). See Section 9.

Gradient-boosted ensembles (2016–2024): supervised meta-predictors

Tool (year)	Algorithm	Pros	Cons
REVEL (2016)	Random forest ensemble of 13 prediction tools	Very high reported AUC ($\sim$0.95); widely adopted	Inputs ClinVar-trained $\Rightarrow$ circular when benchmarked on ClinVar
CADD (2019)	Logistic/RF on simulated negatives + observed positives	Single model for all variant types (incl. non-coding)	Scores uncalibrated; training labels are simulated
VARITY (2021)	XGBoost with noise-weighted training	Handles label noise explicitly	Gene-level split only (no paralog protection)
MAGPIE (2024)	Gradient boosting on multi-omics features	Recent, multi-modal feature space	Paralog-split status undocumented in the publication

Why we ban these as features

`verify_no_leakage.py` excludes REVEL, CADD, ClinPred, M-CAP, MetaSVM — using meta-predictors trained on ClinVar would inflate PR-AUC by $\sim$0.07.

Protein language models (2021–2023): unsupervised representations

Tool (year)	Algorithm	Pros	Cons
EVE (2021)	Variational auto-encoder, one per gene (∼3,000 genes)	Gene-specific deep model; captures epistasis	Limited gene coverage; slow per-gene training
ESM-1b / ESM-2 (2023, zero-shot)	Masked-AA log-likelihood ratio from 250M-sequence corpus	**Fully unsupervised** ⇒ uncontaminated by ClinVar	Lower AUC at small scale (we use `esm2_t12_35M`)
AlphaMissense (2023)	Structural PLM calibrated against ClinVar at release	SOTA reported AUC (∼0.96 on own benchmark)	Calibrated on ClinVar ⇒ comparison contamination on ClinVar tests

The PLM choice in our pipeline

ESM-2 is the only one provably unsupervised ⇒ we integrate its LLR as a Phase-2.1 feature (Section 8). Lift on internal test: ΔPR-AUC $= +0.0313$, $p < 10^{-4}$.

Our contribution + technology stack

Six contributions

1. **Leakage audit** — PR-AUC 0.955 → 0.836 raw / 0.827 calibrated, three named contamination sources removed.
2. **Like-for-like baseline re-scoring** — SIFT, PolyPhen-2, AlphaMissense scored on *our own* paralog-disjoint test (own-coverage CIs).
3. **External validation on denovo-db** — paired bootstrap on ΔROC ($+0.076$, $p = 0.073$) post-constraint vs. pre-constraint.
4. **ESM-2 dual integration** — Phase 1 zero-shot rank fusion + Phase 2.1 training-time feature (ΔPR-AUC $+0.0313$, $p < 10^{-4}$).
5. **Auditable methodology** — 95 % bootstrap CI on every metric, Murphy decomposition, paired tests.
6. **Reproducible artifact** — 157 tests, CI-enforced leakage gate, Docker image, Streamlit demo, LaTeX academic report.

Technology stack

ML core: XGBoost 2.x, Optuna 4 (TPE, 40 trials), SHAP, `scikit-learn`

Protein language model: HuggingFace `transformers`, `esm2_t12_35M_UR50D` (35M params)

Data + features: ClinVar, gnomAD v2.1 + v4, dbNSFP 5.3.1a; `pandas`, `polars`, parquet/pyarrow

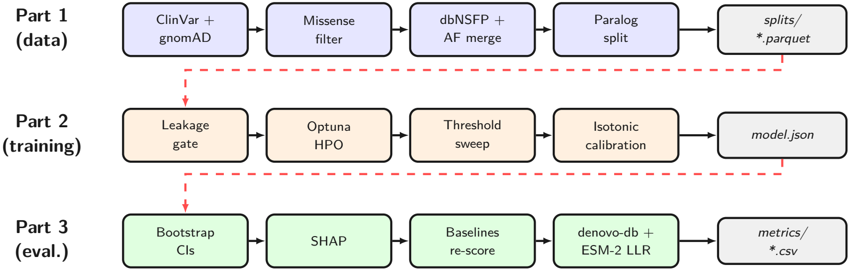
External APIs: Ensembl VEP REST (with on-disk cache fallback)

Infrastructure: Docker (TeX Live 2026), GitHub Actions CI, `pytest` 157, `ruff`, `black`, `mypy`, `pre-commit`

Delivery surfaces: Streamlit demo, Typer CLI, 10 narrative Jupyter notebooks, LaTeX defense + thesis

`github.com/RayanAlDwlah/Genetic-Mutation-Detection-project`

Pipeline overview — three parts, three lanes



Dashed red arrows = inter-part hand-offs (Part 1 splits feed Part 2; Part 2 model feeds Part 3).

Part 1 — Data preprocessing

Sources (joined on GRCh38 `chr:pos:ref:alt`)

- ClinVar — labels (review star ≥ 2)
- gnomAD v2.1 + v4 — AF, AC, AN
- dbNSFP 5.3 — conservation, AA chemistry, Pfam
- gnomAD v2.1.1 — gene-level constraint (pLI, LOEUF, mis_z)

Output 195,098 missense variants → Part 2 as `splits/{train,val,test}.parquet`.

Two critical filters

Missense-only

Drop rows with null `ref_aa` OR null `alt_aa` ⇒ remove ~**88K non-missense rows**. Pre-audit: 64% of pathogenic rows had null AA ⇒ model learned "null ⇒ pathogenic". Fix alone drops PR-AUC 0.955 → 0.819.

Paralog-aware split

Family-level via `assign_gene_family` (16 regex + trailing-digit fallback). **15,479 genes** → **7,851 families**, zero shared across train/val/test (CI-enforced).

Part 2 — Training, leakage audit, calibration

Leakage gate (6 CI-enforced invariants)

1. No banned features (`is_common`, `chr`, `ref`, `alt`)
2. 100 % missense-only rows in training
3. Zero shared gene families (gene + family disjoint)
4. Pathogenic-rate gap < 0.08 across splits
5. Phase-2.1 ESM-2 features pass banned-upstream-meta check
6. ESM-2 split integrity (no test/val overlap in score parquet)

Every push to `main` runs all six; failure blocks the merge.

Five-stage leakage journey (raw PR-AUC)
$0.955 \to 0.819 \to 0.816 \to 0.835 \to 0.836$. Calibrated headline: 0.827.

Model XGBoost + Optuna TPE (multivariate grouping), 40 trials, PR-AUC objective, median pruner after 200 rounds.

Calibration Isotonic on validation; Murphy:
$\text{Brier} = \text{Rel} - \text{Res} + \text{Unc}$.

	Brier	Rel.	ECE
Raw	0.088	0.0054	0.054
Platt	0.083	0.0011	0.029
Isotonic	**0.083**	**0.0002**	**0.011**

Reliability drops 23×; resolution unchanged ⇒ discrimination preserved.

Part 3 — Evaluation framework

Internal evaluation

- Paralog-disjoint test split — **28,098 variants**.
- 95 % bootstrap CI on every metric (1,000 reps, seed 42): ROC-AUC, PR-AUC, F1, MCC, Brier.
- Murphy decomposition for calibration (Reliability, Resolution, Uncertainty).
- **Paired bootstrap on Δ** for ablations ($n = 28{,}098$, same resampled indices for both models).

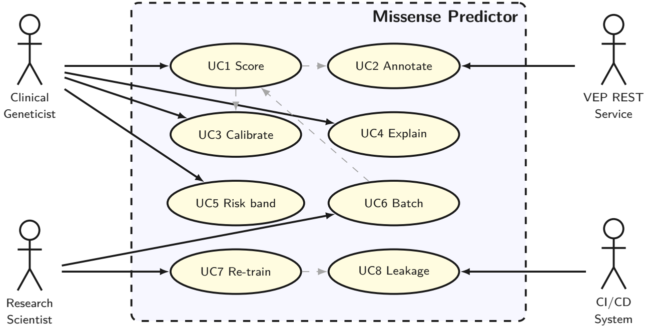
External evaluation

- DENOVO-DB family-holdout — **n = 201** unseen gene families.
- Pre- vs. post-constraint paired bootstrap (recovered checkpoint at `f8ab464`).
- Permutation test for AUC > 0.5 (10,000 reps, seed 42).
- Calibration transfer audit: does isotonic from ClinVar val transfer to denovo-db?

Why this framework

Internal CIs guard against single-split optimism; paired tests guard against unfair Δ comparisons; external denovo-db guards against ClinVar-shaped overfit.

Use case diagram + requirements summary



Functional ($\times 8$)

Ingestion, feature assembly, scoring, risk categorisation, feature attribution, batch mode, external reproduction, report regeneration.

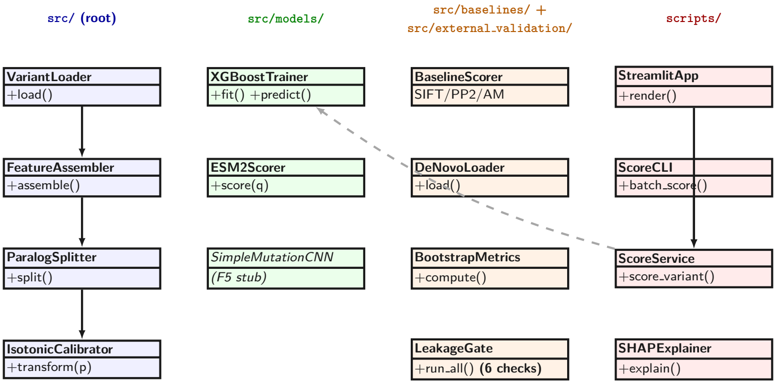
Non-functional ($\times 7$)

Performance (<100 ms cache hit), reliability (10^{-3} seed-reproducible), scalability (10^{6} variants), usability (non-ML user), maintainability (CI), security (no PHI), reproducibility (Docker).

Traceability

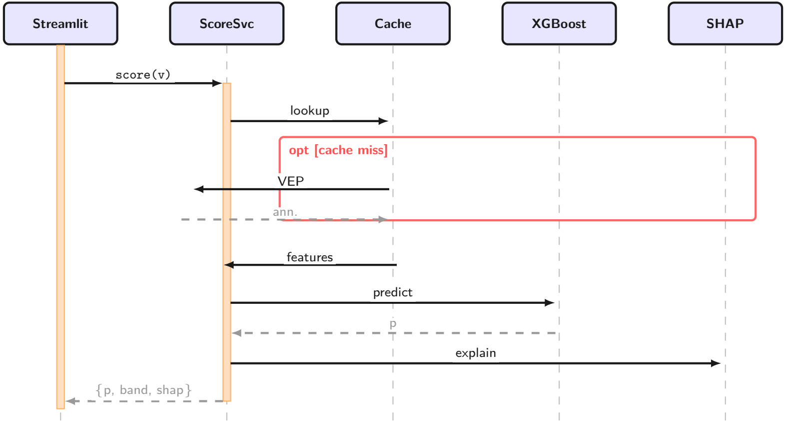
Every requirement maps to ≥ 1 `src/` module + ≥ 1 pytest test case.

Class diagram — four *real* packages from `src/`



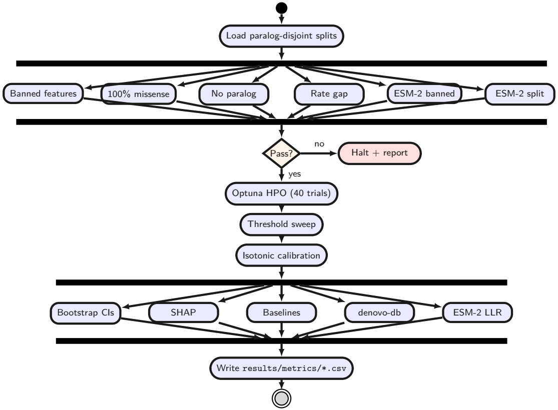
Conceptual blueprint. Production code uses module-level functions in `src/*` rather than the named classes shown here, following Python convention for ML pipelines.

Sequence diagram — single scoring request



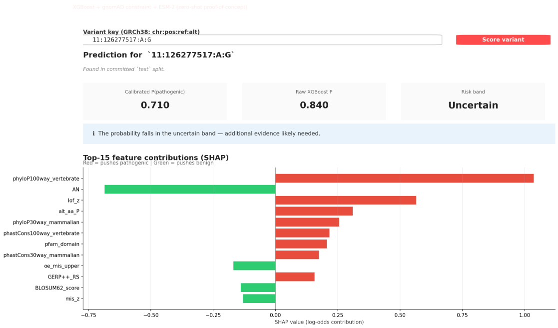
Conceptual blueprint. ScoreSvc, Cache, XGBoost, SHAP map to module-level functions in `scripts/streamlit_app.py` + Pickle/Parquet I/O, not concrete classes.

Activity diagram — training + evaluation workflow



Fork-1 = 6 leakage checks (parallel); Fork-2 = 5 eval tracks. Conceptual flow; production uses sequential scripts. State + Component → appendix.

Interface — Streamlit demo + Typer CLI



Streamlit web demo

Input	GRCh38 variant key `chr:pos:ref:alt`
Output	calibrated p + raw XGBoost + risk band
Explainer	Top-15 SHAP, colour-coded by direction
Latency	$<$100 ms cache hit; $\sim$2 s cache miss

Typer CLI

Single	`score chr:pos:ref:alt`
Batch	`score-batch <vcf|csv>`
Explain	`explain --top 15 <variant>`
Re-train	`make train` (Optuna 40 trials)

Implementation — hardware, software stack, reproducibility

Hardware

- Apple Silicon laptop (MPS GPU)
- Free-tier Colab T4 (ESM-2 scoring only)
- GitHub Actions `ubuntu-22.04`

Software stack

- Python 3.11 + pinned deps
- XGBoost 2.x + Optuna 4
- SHAP, scikit-learn, scipy
- HF `transformers`, PyTorch
- `esm2_t12_35M_UR50D`
- pandas, polars, parquet, pyarrow

Reproducibility

- Docker (TeX Live 2026)
- `requirements-lock.txt`
- Fixed seeds (42)
- **157 pytest tests**, 59% coverage
- CI: ruff + black + mypy + leakage gate
- Snapshot tests on metric files

Single-command reproduction

```
docker build -t missense-classifier . &&
  docker run --rm missense-classifier make reproduce-headline
```

Reproduces every metric within 10^{-3} *on a clean clone.*

Headline results — paralog-disjoint test split

Test split
Paralog-disjoint
28,098 variants
Pathogenic rate $\sim$30%

All 95 % CIs from 1,000-replicate nonparametric bootstrap (seed = 42).

Metric	Value (95 % CI)
ROC-AUC	0.938 [0.935, 0.941]
PR-AUC (uncalibrated)	0.836 [0.827, 0.843]
PR-AUC (calibrated)	**0.827** [0.819, 0.835]
F1	0.775
Brier (calibrated)	0.083
ECE (calibrated)	0.011

Calibration trades a small ranking penalty (ΔPR-AUC $= -0.008$, 0.8355 $\rightarrow$ 0.8273) for $\sim$5$\times$ better probability reliability (ECE 0.054 $\rightarrow$ 0.011). Calibrated probabilities are the headline; uncalibrated is shown for transparency.

External validation — denovo-db

Highlighted rows: holdout from gene families completely unseen in training.

Slice	n	ROC-AUC (95 % CI)	PR-AUC (95 % CI)
full (pre-constraint)	642	0.468 [0.42, 0.52]	0.761
full (post-constraint)	642	**0.511** [0.46, 0.56]	**0.790**
holdout (pre-constraint)	201	0.487 [0.38, 0.58]	0.789
holdout (post-constraint)	201	**0.573** [0.48, 0.67]	**0.838**

The headline

On family-holdout ($n = 201$): paired bootstrap ΔROC $= +0.076$ $[-0.021, +0.172]$, one-sided $p = 0.073$. **Partial closure** of the generalisation gap, not a deployable predictor.

C1 — near-base-rate caveat

Holdout class balance is **80 % positive**.
Constant-positive baseline PR-AUC ≈ 0.80.
Meaningful PR-AUC lift is only $+0.037$, so ROC-AUC is the primary metric for this cohort.

ESM-2 zero-shot + rank fusion

Test split (28,098 variants, paralog-disjoint)

Model	ROC-AUC	PR-AUC	α
XGBoost (calibrated)	0.938	0.827	—
ESM-2 LLR (zero-shot)	0.735	0.604	—
Rank fusion, uniform	0.895	0.784	0.50
Rank fusion, tuned	**0.941**	**0.852**	**0.175**

$$s_i^{\text{fused}} = (1-\alpha)\cdot\frac{r_i^{\text{XGB}}}{n} + \alpha\cdot\frac{r_i^{\text{ESM}}}{n}, \qquad \alpha^* = 0.175$$

Why fusion works

Tuned $\alpha = 0.175$ down-weights the weaker PLM signal. Uniform $\alpha = 0.5$ is *worse* than XGBoost alone. Tuned mix beats XGBoost on internal test (paired DeLong on denovo-db: n.s. at the 5 % level).

Phase 1 takeaway

ESM-2 zero-shot LLR is fully unsupervised — provably uncontaminated by ClinVar. Fusion gives +0.025 PR-AUC lift over XGBoost, motivating training-time integration (Slide 21).

Phase 2.1 — training-time ESM-2 integration + SHAP shift

Ablation (paralog-disjoint test, $n = 28{,}098$)

Model	ROC	PR
Phase-1 baseline	0.938	0.827
Augmented `no_esm2`	0.939	0.840
Augmented `esm2_only`	0.739	0.610
Augmented `full`	**0.948**	**0.865**

Paired bootstrap full vs. no_esm2:
ΔPR-AUC $= +0.0313$ $[+0.0266, +0.0361]$, $p < 10^{-4}$.
ΔROC-AUC $= +0.0093$ $[+0.0082, +0.0103]$, $p < 10^{-4}$.

SHAP shift after ESM-2 added

- `is_imputed_esm2_llr` ranks #1 (mean |SHAP| = 1.05)
- `esm2_llr` ranks #4 (= 0.42)
- Conservation features (phyloP, GERP) remain top-tier; ESM-2 partly substitutes (Spearman $\rho \leq |0.31|$), partly supplements.
- `lof_z` drops from rank 3 (Phase-1) to rank 5.

denovo-db holdout: ΔROC $= -0.13$ + C2 constraint-coverage caveat

The in-distribution gain (+0.031 PR-AUC) does *not* transfer to unseen families. **Partly a coverage artefact (C2):** held-out family genes receive *imputed* constraint values for the augmented model where the baseline had real gnomAD values. Full slice (preserved coverage) shows no degradation — supporting the artefact interpretation, not a pure ESM-2 effect.

Comparison with published methods

All baselines re-scored on our paralog-disjoint test split (28,098 variants).

Method	Year	Methodology	Coverage	PR-AUC
SIFT	2003	legacy, no guards	96%	0.620
PolyPhen-2[†]	2010	legacy, no guards	93%	0.728
AlphaMissense[*]	2023	high ClinVar risk	86%	0.890
ESM-2 LLR (zero-shot)	2023	no supervision	100%	0.604
XGBoost (ours, baseline)	2026	paralog + audit	100%	0.827
XGBoost + ESM-2 LLR (ours)	**2026**	**paralog + audit**	**100%**	**0.865**

What only *we* report

- Paralog-family-disjoint split (CI-enforced)
- Three-source leakage audit
- External validation on DENOVO-DB
- Bootstrap 95 % CIs + paired bootstrap on Δ
- Calibration deep-dive (ECE = 0.011)
- Docker + lock file: 6-decimal reproduction

Honest framing

- +0.245 PR-AUC over SIFT
- +0.137 PR-AUC over PolyPhen-2
- AM gap: -0.063 (Phase-1) $\rightarrow$ $-$**0.025** (Phase-2.1)
- AM is ClinVar-calibrated; matched-coverage on $\mathcal{S}_{\text{AM}}$ shrinks gap further (D7).

Each row on its baseline's own-coverage subset — rows NOT directly comparable.

[*] AM ClinVar-calibrated $\Rightarrow$ test overlaps training signal. [†] PolyPhen-2 trained on HumDiv/HumVar (overlaps ClinVar). REVEL omitted (13 ClinVar-trained base learners).

Future work

1. **Larger-checkpoint ESM-2.** Upgrade to `esm2_t33_650M_UR50D` ($19\times$ parameters); expected to lift the OOD ceiling (current 35M shows ΔROC $= -0.13$ on unseen gene families).
2. **AlphaFold-2 structural features.** pLDDT, SASA, DSSP secondary structure, residue depth — orthogonal to PLM signal, complementary to conservation.
3. **ProteinGym DMS external validation.** Third dataset independent of ClinVar and DENOVO-DB; deep mutational scanning fitness measurements as both validation set and auxiliary training signal.
4. **Hybrid stacking + ACMG integration.** Meta-learner combining XGBoost + ESM-2 + AlphaMissense, packaged as an ACMG-compatible plug-in for clinical workflows.

Deficiencies (D1–D8)

Honest limitations, each paired with a next-step item on Slide 23.

ID	Limitation	Status
D1	External ROC 0.573, CI covers chance ($n = 201$)	open
D2	ClinVar-only training, no DMS labels	open
D3	Isotonic-only calibration, no per-gene	open
D4	ESM-2 zero-shot only on denovo-db	**shipped** (Phase 2.1)
D5	No AlphaFold-2 structural features yet	open
D6	Streamlit-only — no REST API, no VEP plug-in	open
D7	Own-coverage baselines, no matched-coverage rerun on $\mathcal{S}_{\text{AM}}$	open
D8	Pre-constraint preds not persisted	**recovered** (`f8ab464`)

Two of the eight (D4, D8) were resolved during this project. The remaining six map to the four future-work items on Slide 22.

Conclusion — take-home messages

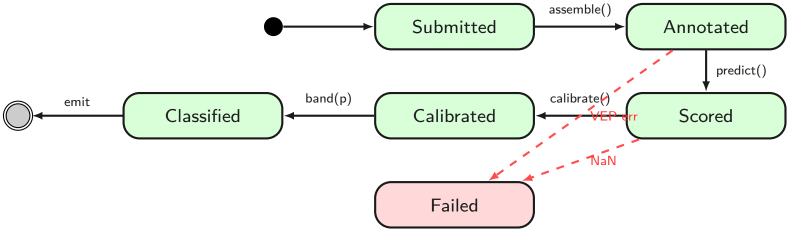
1. **Audited numbers beat inflated ones.** Calibrated PR-AUC 0.827 has no known contamination; the pre-audit 0.955 had three.
2. **Like-for-like baselines change the conversation.** On the same paralog-disjoint test we beat classical tools cleanly (SIFT 0.620, PolyPhen-2 0.728) and sit just below AlphaMissense (-0.025 PR-AUC) under stricter methodology.
3. **Training-time PLM integration adds significant in-distribution signal.** The ESM-2-augmented model lifts test PR-AUC $0.827 \rightarrow 0.865$ ($\Delta = +0.031$, $p < 10^{-4}$, paired bootstrap on $n = 28{,}098$).
4. **External validation is the real benchmark.** Partial closure on DENOVO-DB (paired ΔROC $= +0.076$, CI still covers chance at $n = 201$); the in-distribution ESM-2 gain does *not* transfer to unseen gene families (ΔROC $= -0.13$, motivates a larger PLM checkpoint).

Thank you

Questions?

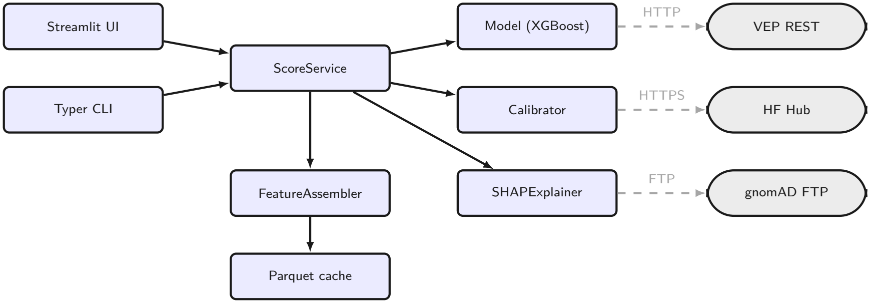
Code + thesis + slides	`github.com/RayanAlDwlah/Genetic-Mutation-Detection-project`
Reproduce headline	`docker run --rm missense-classifier make reproduce-headline`
Streamlit demo	`streamlit run scripts/streamlit_app.py`

Backup — State diagram (variant life-cycle)



Conceptual blueprint of the per-variant life-cycle as implemented in `scripts/streamlit_app.py`*: featurise (VEP or cache) → predict_proba → isotonic.transform → classify by threshold. Try-except handles VEP/NaN errors.*

Backup — Component diagram



Conceptual blueprint. Solid arrows = direct uses; dashed arrows = external network dependencies (VEP HTTP, HF Hub HTTPS, gnomAD FTP). Production code uses module-level functions, not concrete ScoreService/FeatureAssembler classes.

Backup — System architecture (6-layer)

Quality gates

`ruff`
`mypy`
`pytest`
leakage
CI

L6 Presentation — Streamlit ▪ Typer CLI ▪ Python API

L5 Application — validation ▪ cache ▪ payload

L4 Service — score_variant() ▪ explain() ▪ batch_score()

L3 ML core — XGBoost ▪ isotonic ▪ SHAP ▪ ESM-2 LLR

L2 Data access — pyarrow ▪ paralog splitter ▪ assembler

L1 Data — ClinVar ▪ gnomAD ▪ dbNSFP ▪ DENOVO-DB ▪ UniProt

Repro-ducibility

Docker
lock file
fixed seed
snapshots

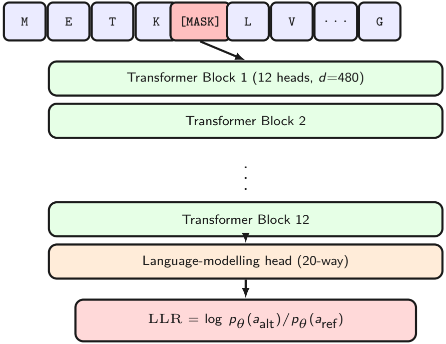
Backup — Calibration triptych + Murphy decomposition

Murphy: Brier = Reliability − Resolution + Uncertainty

Calibrator	Brier	Reliability	Resolution	ECE	Log-loss
Raw	0.0876	0.00543	0.1055	0.054	0.286
Platt	0.0834	0.00114	0.1055	0.029	0.278
Isotonic	**0.0826**	**0.00024**	0.1053	**0.011**	**0.273**

- Resolution constant ⇒ discrimination unchanged (post-hoc monotone calibration cannot change ranking).
- Reliability drops 23× under isotonic ⇒ probability calibration works.
- ECE 0.054 → 0.011 < 0.02 clinical target.

Backup — ESM-2 zero-shot scoring architecture



Model: `facebook/esm2_t12_35M_UR50D`

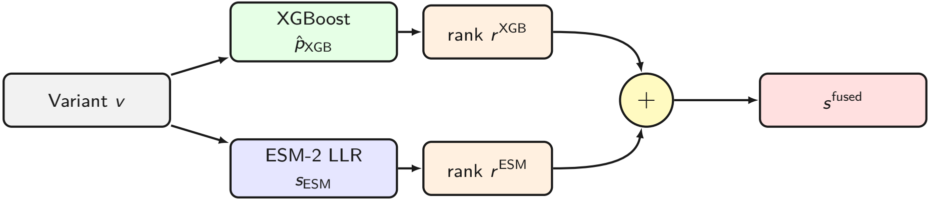
- 12 transformer blocks
- $d = 480$, 20 attention heads
- ∼35M parameters
- **Zero-shot**: no weight updates

Log-likelihood ratio:

$$\mathrm{LLR}_{\text{ESM-2}} = \log \frac{\mathbb{P}_\theta(a_{\text{alt}} \mid s_{\setminus \pi})}{\mathbb{P}_\theta(a_{\text{ref}} \mid s_{\setminus \pi})}$$

where $s_{\setminus \pi}$ is the sequence with position π masked.
Lower LLR $\Rightarrow$ more pathogenic.

Backup — Rank fusion operator



$$s_i^{\mathrm{fused}} = (1-\alpha)\cdot\frac{r_i^{\mathrm{XGB}}}{n} + \alpha\cdot\frac{r_i^{\mathrm{ESM}}}{n}, \qquad \alpha^* = 0.175$$

Convention matches `scripts/run_rank_fusion_esm2.py:99-100`: α is the weight on ESM-2. Tuned $\alpha^* = 0.175$ down-weights the weaker zero-shot signal; uniform $\alpha = 0.5$ is worse than XGBoost alone.

Backup — denovo-db: paired bootstrap vs permutation test

Two separate tests — different null hypotheses

- **Permutation test** ($n_{\text{perm}} = 10{,}000$, seed 42):
  Null: $\text{AUC}_{\text{post}} \leq 0.5$ ($p_{\text{perm}} = 0.0744$).
  Tests *"is the post-constraint score better than chance?"*
- **Paired bootstrap** ($n_{\text{boot}} = 1{,}000$, seed 42, recovered checkpoint at `f8ab464`):
  Null: $\Delta\text{ROC} = \text{AUC}_{\text{post}} - \text{AUC}_{\text{pre}} \leq 0$ ($p_{\text{paired}} = 0.073$).
  Tests *"did adding constraint features help, on the same 201 variants?"*

Slice	Pre-constraint	Post-constraint	Paired Δ
full ROC ($n = 642$)	0.468	**0.511**	+0.043
full PR ($n = 642$)	0.761	**0.790**	+0.029
holdout ROC ($n = 201$)	0.487	**0.573**	**+0.0764** [−0.021, +0.172]
holdout PR ($n = 201$)	0.789	**0.838**	+0.0375 [−0.005, +0.080]

Both p-values round to ∼0.07 by coincidence; they answer different questions. The defense headline (Slide 19) uses the paired bootstrap result — the right test for a Δ claim.

Backup — CI: 157 tests on every push to `main`

```
$ pytest tests/ --tb=no -q

................................................................ [ 40%]
................................................................ [ 81%]
.................................s..                             [100%]

Required test coverage of 55% reached. Total coverage: 59.07%

SKIPPED [1] tests/integration/test_reproduce_headline.py:83:
  missing: results/metrics/xgboost_predictions.parquet

============ 157 passed, 1 skipped in 12.28s ============
```

Five-level test pyramid: Unit (120 tests, <10 s), Property (8 `hypothesis`), Integration (25, 1–2 min), Scientific / E2E (5 scripts, 5–15 min), CI gate (blocks merge on red). Single skip is the GPU-only ESM-2 reproduction smoke test.